Source Verification and Final Review

Practice exams and assessment validity in computer science

27 October 2026

This record documents source verification and an internal critical review of the final paper's 21 references. It distinguishes reported findings, interpretations, and policy recommendations, and records the evidence and limitations supporting each claim group. The internal review considered measurement validity, the case for instructor discretion, and academic presentation. It does not represent external journal peer review.

What was checked

The review compared claims with primary publications, author manuscripts, official webpages, and publisher or institutional bibliographic records. It checked author order and initials, publication year, titles, proceedings or journal information, page ranges, identifiers, sample sizes, effect sizes, comparison groups, and the distinction between findings and interpretation. Publication-year citations follow the published issue, rather than a later archive upload or an earlier online-first date.

For 19 of the 21 entries, full primary article or report text, or the complete official webpage, was examined. Arendasy et al. (2016) and Messick (1981) were checked against publisher or official abstracts and bibliographic records. Their use is confined to claims those records explicitly support. The audit checks how the paper represents the cited evidence; it does not independently reanalyse the datasets or every study included in a meta-analysis.

Final review decisions

- Practice testing, an extended practice examination, and a full mock examination are defined separately. General retrieval-practice findings are not treated as estimates of the effect of a timed simulation.
- The recommendation states its educational premise and allows relevant prerequisites, varied preparation, and legitimate instructor judgments about unnecessary mock-exam features. It does not require an analogue of every assessed question.
- Direct computing evidence is identified as introductory or novice programming research. Extension to other CS subjects is explicitly a reasoned application. Voluntary uptake in Fossati and the already-prepared comparison groups in Smith are distinguished from availability effects.
- LSAT estimates are labelled as adjusted overall comparisons. Statistical contrasts and subgroup limits are stated.
- Study descriptions retain the verified experimental-arm distinctions, outcome boundaries, and observational limits recorded below. The Bloom examples remain illustrative. The search approach and two abstract-based checks are disclosed.

- All 21 references have corresponding uses in the paper. Author names, dates, titles, journal or proceedings details, identifiers, and source locations were checked. Table numbering and cross-references were reconciled, and the rendered documents were visually reviewed.

Source verification register

Locators below refer to printed publication pages unless identified as a section or table. Links lead to the checked primary text or official record. The full bibliographic entries appear in the revised paper. A verified representation of a study does not make its design causal or eliminate its limitations.

Reference	Primary material	Locator	Verification result
Agarwal (2019)	Full article	pp. 193 and 196–203	Higher-order practice findings verified. Population and experimental-arm distinctions corrected.
AERA APA NCME (2014)	Full standards	pp. 2 and 7; Standards 4.13–4.16 pp. 90–91	Conditional guidance verified; scope for classroom tests clarified.
Arendasy et al. (2016)	Publisher abstract and metadata	Abstract	N = 1,768 and individual-subtest invariance verified. Observational, outside CS; full methods not audited.
Biggs (1996)	Full article	pp. 350–351 and 360–361	Alignment claim verified. Conceptual source; not a practice-exam intervention. Issue 32(3) supplied.
Butler et al. (2017)	Full article	pp. 435–441	Initial closed-book transfer findings verified; Experiment 4 reanswer boundary added.
Dustman et al. (2021)	Full official report	pp. 14–15, Tables 7–8; pp. 17–18	Category sequence verified; comparison group and inferential limits stated.
Fossati and Hashemi Tonekaboni (2020)	Full one-page poster paper	p. 1362, Methods and Results	Association verified. Practice was available to all; comparison concerns uptake. Self-selection limits causal interpretation.
Krathwohl (2002)	Full article	pp. 213–215 and 217–218	Dimensions, categories, overlapping hierarchy, and alignment use verified. Taxonomy is not intervention evidence.
Krinsky (2024)	Official LSAC webpage	Opening two paragraphs	Added source verifies removal of Analytical Reasoning beginning August 2024.

Reference	Primary material	Locator	Verification result
LSAC (n.d.)	Official practice-test webpage	Opening guidance before LawHub section	Advice about mechanics, instructions, question types, and pacing verified. Guidance is not an experiment.
Messick (1981)	Official report abstract and metadata	ETS abstract; RR-81-19	Three preparation mechanisms verified from abstract. The report-format citation and 1981 date are correct.
Messick (1995)	Full article	pp. 741–745	Score interpretation, underrepresentation, irrelevant difficulty, and irrelevant easiness verified. CS examples are applications.
Öqvist and Nouri (2018)	Full article	Methods Results and Discussion	N = 20 and nonsignificant overall result verified. One task differed; equivalence is not established. Issue 5(2) supplied.
Pan and Rickard (2018)	Full article manuscript	Overall Meta-Analyses; Table 4	122 experiments, N = 10,382, d = 0.40, CI [0.31, 0.50] verified. Conditional transfer caveats retained.
Smith et al. (2023)	Full article	pp. 758–760; Tables 1–2	Specified comparison results verified. Both cohorts had practice exams; nonsignificant programming contrasts are identified.
Thompson et al. (2008)	Full author text and institutional record	pp. 155–161	Classification argument and metadata verified. Direct ACM catalogue URL retained; no Crossref record was returned for its identifier.
UW Tacoma (n.d.)	Official advising webpage	How important is it to take full practice exams	15–20 recommendation verified as advising guidance. Its components section still describes the older LSAT structure.
van Gog et al. (2015)	Full article	Abstract; pp. 283–285	Four experiments and lack of delayed advantage verified. No-feedback and short-instruction context limits extrapolation. Issue 27(2) supplied.
Wu and Smith (2024)	Full article manuscript	Sections 4.1 6.3 and 7	Related-category comparison verified; no matched-question equivalence claim. Exact proceedings title corrected.
Yang et al. (2023)	Full accepted manuscript	Results and footnotes	24 studies, N = 3,374, g = −0.52 verified. Pooled outcome wording refined; issue 35(3) supplied.

Reference	Primary material	Locator	Verification result
Yang et al. (2021)	Full typeset proof and published record	Results; Questions 4–7	222 studies, N = 48,478, g = 0.499 verified. Heterogeneous classroom interventions; no isolated mock-exam effect.

Interpretive limits

The final paper presents a reasoned policy recommendation. The reviewed studies do not establish a universal duty to distribute a complete practice paper or full mock, or an inherent learning benefit from withholding representative preparation. The recommendation therefore retains flexibility about form, relevant prerequisites, and substantive novelty. Judging a particular course requires assessment-specific evidence. Source verification supports faithful representation of the cited literature and cannot remove the limitations of the underlying studies.